Network Exposure and Traffic Monitoring for Security Analysis

Project Report Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

By

B. Kundana Mani (2420030558)

Potnuru Bhavya Sree (2420030506)

Under the Esteemed Guidance of

Mr. Shaik Asif

Of

Section: 5

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

ABSTRACT

With the rapid growth of networked systems, ensuring secure communication and minimising network exposure have become essential aspects of cybersecurity. Many security incidents occur due to exposed network ports, misconfigured services, or insecure data transmission, often without the awareness of system administrators. Understanding how systems respond to network requests and how data flows across a network is a fundamental requirement for identifying potential security risks.

This project presents a combined cybersecurity assessment approach by integrating network scanning, firewall behaviour analysis, and network traffic analysis. Network scanning is performed using Nmap to identify open, closed, and filtered ports, which helps assess service exposure and infer firewall or packet-filtering mechanisms. Additionally, network traffic analysis is conducted to observe real-time communication patterns, identify commonly used protocols, and evaluate whether data transmission is encrypted or unencrypted.

The results of this project demonstrate how basic security assessment techniques can provide valuable insights into a system's network posture. By correlating port states, firewall behaviour, and traffic characteristics, the project highlights the importance of minimising attack surface and ensuring secure communication. This combined approach helps beginners understand real-world cybersecurity concepts while following ethical and non-intrusive assessment practices.

PROBLEM STATEMENT

Modern computer systems are continuously connected to local and external networks, increasing their exposure to potential cyber threats. Many systems operate with limited visibility into which network ports are accessible, which services are running, and whether network traffic is adequately protected. Open or improperly managed ports can serve as entry points for attackers, while insecure or unmonitored network traffic can lead to data interception and privacy risks.

One of the primary challenges in cybersecurity is identifying network exposure and communication behaviour without performing intrusive or exploitative actions. Attackers often begin with reconnaissance techniques such as port scanning and traffic observation to understand system behaviour. However, system administrators and students often lack practical knowledge of how these techniques work and how the results should be interpreted from a defensive security perspective.

The problem addressed by this project is the lack of a structured and beginner-friendly approach to assessing network security using ethical methods. This project aims to identify port states, infer firewall behaviour through filtered responses, and analyse network traffic patterns to determine whether communication is secure. By combining network scanning and traffic analysis into a single assessment, the project provides a comprehensive view of network exposure and communication security, helping users understand how basic cybersecurity assessments contribute to stronger system protection.

SOFTWARE REQUIREMENTS

- Operating System: Kali Linux
- Network Scanning Tool: Nmap
- Terminal Emulator: Default Kali Linux Terminal
- Virtualisation Software (Optional): VirtualBox / VMware
- Documentation Tools: LibreOffice / MS Word
- Web Browser: Firefox (for research and documentation)